

Practice Problems List-11 (Recursion)

Solve the following problems **using recursion**:

1. Write a C/C++ program to Print 1 to 10.
2. Write a C/C++ program to Input a and b using scanf. Print a to b.
3. Write a C/C++ program to Print 100 to 1.
4. Write a C/C++ program to Print the even numbers between 1 and 100.
5. Write a C/C++ program to Print the following pattern:

```
1
2
3
4
5
5
4
3
2
1
```

6. Write a C/C++ program to Print the sum of 1 to 10.
7. Write a C/C++ program to Calculate the sum of the following series:
$$1^2 + 2^2 + 3^2 + 4^2 + 5^2$$
8. Write a C/C++ program to Write a program to print the factorial of a number.
9. Write a C/C++ program to Take input two integers x and y and calculate x^y .
10. Write a C program to print all the divisors/factors of a number.
11. Write a C program to check whether a number is prime or not.
12. Write a C program to calculate the GCD/HCF of two numbers.
13. Write a C program to calculate the LCM of two numbers.
14. Write a C program to count the number of digits in a number.
15. Write a C program to count the sum of digits in a number.
16. Write a C program to find the length of a *C string* (character array).
17. Write a C program to check if a given string is a palindrome.

18. Write a C program to print the reverse of a given string.
19. Write a recursive function to find the sum of elements in an array.
20. Write a recursive function to print the max-element of an array.